CHAPTER 3

MATRICES

CHAPTER ESSENCE AT A GLANCE:
Matrix algebra, matrix multiplication properties, transpose symmetry, orthogonal matrices, and elementary row operations.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Matrix Multiplication and Non-Commutativity:** Matrix multiplication follows a row-by-column dot product procedure that requires the column count of the first matrix to equal the row count of the second. In general matrix multiplication is non-commutative, meaning product AB does not equal BA.

• **Transpose and the Reversal Law:** The transpose operation swaps rows into columns; taking the transpose of a product reverses the order of multiplication, equating the transpose of AB to the product of the transpose of B and the transpose of A.

• **Symmetric and Skew-Symmetric Decomposition:** Every square matrix can be expressed uniquely as the sum of a symmetric matrix and a skew-symmetric matrix by combining half the sum with half the difference of the matrix and its transpose.

• **Skew-Symmetric Diagonal Zero Property:** Every main diagonal element of a skew-symmetric matrix must equal zero because every diagonal element is constrained to equal its own negative.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$(AB)' = B'A' \quad \text{and} \quad (A+B)' = A' + B'$$

Reversal law of matrix transpose for matrix products

$$A = \tfrac{1}{2}(A + A') + \tfrac{1}{2}(A - A') = P + Q$$

Unique decomposition into symmetric and skew-symmetric parts

$$A \cdot A' = A' \cdot A = I \Leftrightarrow A \text{ is Orthogonal}$$

Condition for an orthogonal square matrix

$$A = -A' \rightarrow a_{ii} = -a_{ii} \rightarrow 2a_{ii} = 0 \rightarrow a_{ii} = 0$$

Vanishing of all main diagonal elements in skew-symmetric matrices

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Verification Requirement] When expressing matrix as sum of symmetric and skew-symmetric, examiners deduct 1 Mark if you do not verify BOTH P' = P and Q' = -Q AND confirm P + Q = A!

• **Alert:** [Trap Alert: Skew Diagonal Proof] To prove diagonal elements of skew-symmetric matrix are zero: $a_{ii} = -a_{ii}$ implies $2a_{ii} = 0$ implies $a_{ii} = 0$.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define: (a) Scalar matrix, (b) Symmetric matrix, (c) Skew-symmetric matrix with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Definitions]** 1. Scalar Matrix: A diagonal matrix whose all diagonal elements are equal (a_{ij} = k for i = j, and a_{ij} = 0 for i ≠ j). Example: [[3, 0], [0, 3]]. 2. Symmetric Matrix: A square matrix A such that A' = A ($a_{ij} = a_{ji}$ for all i, j). Example: [[1, 2], [2, 5]]. 3. Skew-Symmetric Matrix: A square matrix A such that A' = - A ($a_{ij} = - a_{ji}$ for all i, j). All diagonal elements are zero (a_{ii} = 0). Example: [[0, 3], [-3, 0]]. [2 Marks]

• **[Boxed Result: Scalar (equal diagonals), Symmetric (A'=A), Skew-Symmetric (A'=-A)]**

2. Construct a 2 × 2 matrix A = [a_{ij}] whose elements are given by $a_{ij} = (i + 2j)^2$ over 2. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Formula & Element Calculation]** $a_{ij} = (i + 2j)^2$ over 2. Order 2 × 2. $a_{11} = (1 + 2(1))^2$ over $2 = 3^2$ over 2 = 9 over 2. $a_{12} = (1 + 2(2))^2$ over $2 = 5^2$ over 2 = 25 over 2. $a_{21} = (2 + 2(1))^2$ over $2 = 4^2$ over 2 = 16 over 2 = 8. $a_{22} = (2 + 2(2))^2$ over $2 = 6^2$ over 2 = 36 over 2 = 18. [1.5 Marks]

• **[Matrix Output]** A = [[9 over 2, 25 over 2], [8, 18]]. [0.5 Mark]

• **[Boxed Result: A = [[9 over 2, 25 over 2], [8, 18]]]**

3. If A = [[1, 2], [3, 4]] and B = [[2, 0], [1, 3]], find AB and BA. Is AB = BA? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Matrix Multiplication]** A = [[1, 2], [3, 4]], B = [[2, 0], [1, 3]]. AB = [[1(2)+2(1), 1(0)+2(3)], [3(2)+4(1), 3(0)+4(3)]] = [[2+2, 0+6], [6+4, 0+12]] = [[4, 6], [10, 12]]. [1.5 Marks]

• BA = [[2(1)+0(3), 2(2)+0(4)], [1(1)+3(3), 1(2)+3(4)]] = [[2+0, 4+0], [1+9, 2+12]] = [[2, 4], [10, 14]]. [1 Mark]

• Since AB ≠ BA, matrix multiplication is NOT commutative in general. [0.5 Mark]

• **[Boxed Result: AB = [[4, 6], [10, 12]]; BA = [[2, 4], [10, 14]]; AB ≠ BA]**

4. If A and B are symmetric matrices of the same order, prove that AB - BA is a skew-symmetric matrix. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Given & Reversal Law]** A and B are symmetric matrices: A' = A and B' = B. Consider transpose of (AB - BA): (AB - BA)' = (AB)' - (BA)' = B' A' - A' B' (by reversal law of transpose: (XY)' = Y' X'). [1.5 Marks]

• **[Substitution]** Substituting B' = B and A' = A: = B A - A B = - (AB - BA). Since the transpose of (AB - BA) equals its negative, (AB - BA) is a SKEW-SYMMETRIC MATRIX. [1.5 Marks]

• **[Boxed Result: (AB - BA)' = - (AB - BA) ⇒ AB - BA is Skew-Symmetric]**

5. For matrix A = [[1, 5], [6, 7]], verify that (A + A') is symmetric and (A - A') is skew-symmetric. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Compute Transpose]** A = [[1, 5], [6, 7]], A' = [[1, 6], [5, 7]]. [0.5 Mark]

• **[Step 2: Symmetric Part]** A + A' = [[1+1, 5+6], [6+5, 7+7]] = [[2, 11], [11, 14]]. (A + A')' = [[2, 11], [11, 14]] = A + A'. Hence (A + A') is SYMMETRIC. [1 Mark]

• **[Step 3: Skew-Symmetric Part]** A - A' = [[1-1, 5-6], [6-5, 7-7]] = [[0, -1], [1, 0]]. (A - A')' = [[0, 1], [-1, 0]] = - [[0, -1], [1, 0]] = - (A - A'). Hence (A - A') is SKEW-SYMMETRIC. [1.5 Marks]

• **[Boxed Result: (A + A') is Symmetric and (A - A') is Skew-Symmetric]**

6. Prove that any square matrix can be uniquely expressed as the sum of a symmetric and a skew-symmetric matrix. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Theorem Proof]** Any square matrix A can be written as: A = (1 over 2)(A + A') + (1 over 2)(A - A') = P + Q. [1 Mark]

• 1. P is symmetric: P' = [(1 over 2)(A + A')]' = (1 over 2)(A' + (A')') = (1 over 2)(A' + A) = P. 2. Q is skew-symmetric: Q' = [(1 over 2)(A - A')]' = (1 over 2)(A' - (A')') = (1 over 2)(A' - A) = - (1 over 2)(A - A') = - Q. [1.5 Marks]

• 3. Uniqueness: Suppose A = R + S where R is symmetric (R'=R) and S is skew-symmetric (S'=-S). A' = R' + S' = R - S. Adding: A + A' = 2R implies R = (1 over 2)(A + A') = P. Subtracting: A - A' = 2S implies S = (1 over 2)(A - A') = Q. The representation is unique! [1.5 Marks]

• **[Boxed Result: A = P + Q where P = (1 over 2)(A+A') (symmetric) and Q = (1 over 2)(A-A') (skew-symmetric)]**

7. Express matrix A = [[3, 5], [1, -1]] as the sum of a symmetric and a skew-symmetric matrix. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Compute P and Q]** A = [[3, 5], [1, -1]]. A' = [[3, 1], [5, -1]]. P = (1 over 2)(A + A') = (1 over 2) [[3+3, 5+1], [1+5, -1-1]] = (1 over 2) [[6, 6], [6, -2]] = [[3, 3], [3, -1]]. (Symmetric: P' = P). [1.5 Marks]

• Q = (1 over 2)(A - A') = (1 over 2) [[3-3, 5-1], [1-5, -1-(-1)]] = (1 over 2) [[0, 4], [-4, 0]] = [[0, 2], [-2, 0]]. (Skew-symmetric: Q' = -Q). [1.5 Marks]

• **[Step 2: Verification of Sum]** P + Q = [[3+0, 3+2], [3-2, -1+0]] = [[3, 5], [1, -1]] = A. [1 Mark]

• **[Boxed Result: A = [[3, 3], [3, -1]] (symmetric) + [[0, 2], [-2, 0]] (skew-symmetric)]**